

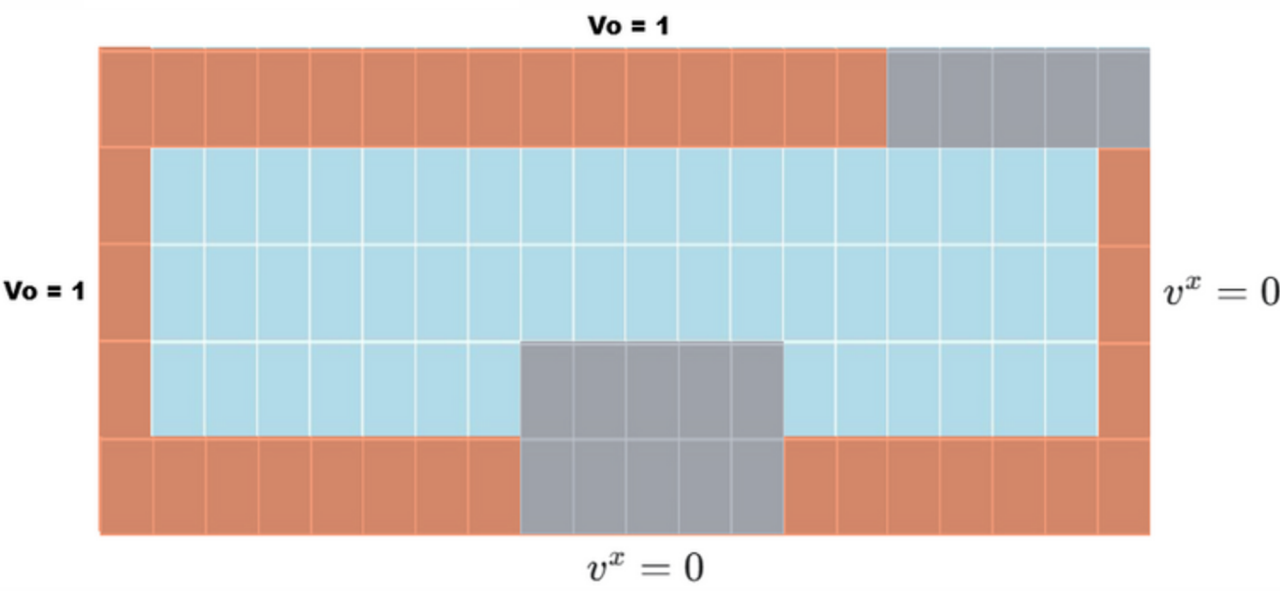
Aplicación del método de Newton-Raphson

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$$i \in [0, 49] \text{ y } j \in [0, 4].$$



2. DEFINICIÓN DEL SISTEMA DE ECUACIONES

$$\mathbf{F}(\mathbf{V}) = \begin{pmatrix} v_{1,1}^x \\ v_{2,1}^x \\ \vdots \\ v_{48,1}^x \\ v_{1,2}^x \\ v_{2,2}^x \\ \vdots \\ v_{48,2}^x \\ v_{1,3}^x \\ v_{2,3}^x \\ \vdots \\ v_{48,3}^x \end{pmatrix}^T$$

144 – 10 = **134** incógnitas.
(Nodos 20 a 29 omitidos)

$$i \in [1, 48] \text{ y } j \in [1, 3].$$

1. FORMULACIÓN DEL SISTEMA

$$v_{i,j}^x = \frac{1}{4} \left\{ v_{i+1,j}^x + v_{i-1,j}^x + v_{i,j+1}^x + v_{i,j-1}^x - 4v_{i,j}^x (v_{i+1,j}^x - v_{i-1,j}^x) - 4v_{i,j}^y (v_{i,j+1}^x - v_{i,j-1}^x) \right\}$$

$$F_{i,j}(\mathbf{V}) = 0. \quad \mathbf{V} = (v^x)$$

$$F_{i,j}(\mathbf{V}) = 4v_{i,j}^x - \left\{ v_{i+1,j}^x + v_{i-1,j}^x + v_{i,j+1}^x + v_{i,j-1}^x - 4v_{i,j}^x (v_{i+1,j}^x - v_{i-1,j}^x) \right\} = 0$$

3. APLICACIÓN DEL ALGORITMO NEWTON-RAPHSON

$$\mathbf{J} = \begin{pmatrix} \frac{\partial F_1}{\partial V_1} & \frac{\partial F_1}{\partial V_2} & \cdots & \frac{\partial F_1}{\partial V_{134}} \\ \frac{\partial F_2}{\partial V_1} & \frac{\partial F_2}{\partial V_2} & \cdots & \frac{\partial F_2}{\partial V_{134}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_{134}}{\partial V_1} & \frac{\partial F_{134}}{\partial V_2} & \cdots & \frac{\partial F_{134}}{\partial V_{134}} \end{pmatrix}$$

$$V^{(k+1)} = V^{(k)} - \underbrace{J(V^{(k)})^{-1} F(V^{(k)})}_{\text{Requiere calcular } J^{-1}}$$

$$J(\mathbf{V}^{(k)}) \cdot \Delta \mathbf{V}^{(k)} = -\mathbf{F}(\mathbf{V}^{(k)})$$

$$\mathbf{V}^{(k+1)} = \mathbf{V}^{(k)} + \Delta \mathbf{V}$$

$$F_{i,j} = 4v_{i,j}^x - v_{i+1,j}^x - v_{i-1,j}^x - v_{i,j+1}^x - v_{i,j-1}^x + 4v_{i,j}^x v_{i-1,j}^x - 4v_{i,j}^x v_{i+1,j}^x$$

DERIVADA - NODO CENTRAL ($v_{i,j}^x$)

$$\frac{\partial F_{i,j}^x}{\partial v_{i,j}^x} = 4 + 4v_{i+1,j}^x - 4v_{i-1,j}^x$$

DERIVADA - V. DERECHO ($v_{i+1,j}^x$)

$$\frac{\partial F_{i,j}^x}{\partial v_{i+1,j}^x} = -1 + 4v_{i,j}^x$$

ALGORITMO QUE PERMITE CONSTRUIR LA MATRIZ JACOBIANA Y EL VALOR RESIDUAL EN CADA ITERACIÓN DEL MÉTODO.

DERIVADA - V. IZQUIERDO ($v_{i-1,j}^x$)

$$\frac{\partial F_{i,j}^x}{\partial v_{i-1,j}^x} = -1 - 4v_{i,j}^x$$

[LINK DEL COLAB](#)

DERIVADA - V. SUPERIOR ($v_{i,j+1}^x$)

$$\frac{\partial F_{i,j}^x}{\partial v_{i,j+1}^x} = \frac{\partial}{\partial v_{i,j+1}^x}(-v_{i,j+1}^x) = -1$$

DERIVADA - V. INFERIOR ($v_{i,j-1}^x$)

$$\frac{\partial F_{i,j}^x}{\partial v_{i,j-1}^x} = \frac{\partial}{\partial v_{i,j-1}^x}(-v_{i,j-1}^x) = -1$$

¡MUCHAS GRACIAS!